SHOPZETU

E-Commerce Customer Analysis

SQL Exploratory Data Analysis | 2023 to 2024

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Tools: SQL Server | 7 Queries | 20 Customers | 52 Orders

NOTE	This analysis covers order data from January 2023 to July 2024 across 20 customers and 52 orders. June and July 2024 data may be incomplete and findings from those months should be verified before acting on them.

Introduction

This report presents a SQL-based exploratory data analysis of ShopZetu, an online fashion and lifestyle store serving customers across Kenya. The analysis was commissioned to answer key business questions around revenue growth, customer retention, product performance, and customer value segmentation.

The analyst was tasked with answering seven specific business questions using structured SQL queries written in SQL Server. Each finding is presented with the business question, the query used, the results returned, and an analytical insight with a recommendation.

Dataset Overview

The ShopZetu database contains four related tables:

Table	Key Fields	Description
Customers	customer_id, full_name, city, join_date	Profile and registration date of each customer
Orders	order_id, customer_id, order_date	Each order placed including date and delivery status
Order Items	order_id, product_id, quantity, unit_price	Line items per order including quantity and price
Products	product_id, product_name, category, unit_price	Product catalogue with categories and pricing

Business Questions Investigated

- Query 1: How has monthly revenue trended from January 2023 to July 2024?
- Query 2: How many new customers joined ShopZetu each month?
- Query 3: Which products generate the highest and lowest revenue?
- Query 4: How is revenue distributed across product categories?
- Query 5: What percentage of customers make repeat purchases?
- Query 6: What is the average number of days between a first and second purchase?
- Query 7: How do customers segment by Recency, Frequency, and Monetary value?

Analyst

Benson Kioko | Data Analytics | 2024

Query 1 Monthly Revenue Trend

Business Question

How has ShopZetu total revenue trended month by month from January 2023 to July 2024? Is the business growing, flat, or declining?

SQL Query

```
SELECT
  FORMAT(O.order_date, 'yyyy-MM') AS OrderMonth,
  SUM(OI.quantity * OI.unit_price) AS TotalRevenue
FROM Order_Items OI
INNER JOIN Orders O ON OI.order_id = O.order_id
GROUP BY FORMAT(O.order_date, 'yyyy-MM')
ORDER BY MIN(O.order_date);
```

Result Set

Month	Total Revenue (KES)
2023-01	10,100
2023-02	7,950
2023-03	9,800
2023-04	11,250
2023-05	15,100
2023-06	19,300
2023-07	16,800
2023-08	20,200
2023-09	**24,150**
2023-10	16,850
2023-11	16,000
2023-12	16,700
2024-01	**24,700**
2024-02	22,100
2024-03	14,600
2024-04	15,500
2024-05	18,000
2024-06	6,600
2024-07	7,700

Insight

> Revenue grew consistently through 2023, peaking at KES 24,700 in January 2024, likely driven by back-to-school and new year spending cycles. It then declined sharply from March 2024, reaching KES 6,600 in

June 2024, the lowest month in the dataset. Before concluding this is a business problem, two things must be verified: whether June and July 2024 data is complete, and whether this reflects a normal mid-year seasonal trough. Full year 2024 data is required before drawing conclusions on the decline.

Query 2 New Customer Acquisition per Month

Business Question

How many new customers joined ShopZetu each month? Is the business growing its customer base over time?

SQL Query

```
SELECT
  FORMAT(join_date, 'yyyy-MM') AS JoiningMonth,
  COUNT(*) AS TotalCustomers
FROM Customers
GROUP BY FORMAT(join_date, 'yyyy-MM')
ORDER BY MIN(join_date);
```

Result Set

Month	New Customers
2023-01	2
2023-02	2
2023-03	2
2023-04	2
2023-05	2
2023-06	2
2023-07	2
2023-08	2
2023-09	2
2023-10	2

Insight

ShopZetu acquired exactly 2 customers per month from January to October 2023, totalling 20 customers, and has acquired no new customers since. Despite this, revenue continued growing into early 2024, confirming that existing customers are increasing their spend over time. However, the complete absence of new customer acquisition since October 2023 creates a concentration risk. Any loss of existing customers cannot be replaced. Recommendation: launch a structured acquisition campaign targeting at least 5 new customers per month to reduce dependency on the existing base.

Query 3 Top 10 and Bottom 10 Products by Revenue

Business Question

Which products generate the most and least revenue for ShopZetu? Show total units sold and total revenue per product.

SQL Query

```
WITH ProductRevenue AS (
  SELECT
    P.product_name,
    P.category,
    SUM(OI.quantity * OI.unit_price) AS TotalRevenue,
    SUM(OI.quantity) AS TotalUnitsSold,
    DENSE_RANK() OVER (ORDER BY SUM(OI.quantity * OI.unit_price) DESC) AS TopRank,
    DENSE_RANK() OVER (ORDER BY SUM(OI.quantity * OI.unit_price) ASC)  AS BottomRank
  FROM Order_Items OI
  INNER JOIN Products P ON OI.product_id = P.product_id
  GROUP BY P.product_name, P.category
)
SELECT product_name, category, TotalRevenue, TotalUnitsSold,
  CASE WHEN TopRank <= 10 THEN 'Top 10'
       WHEN BottomRank <= 10 THEN 'Bottom 10' END AS RankGroup
FROM ProductRevenue
WHERE TopRank <= 10 OR BottomRank <= 10
ORDER BY TotalRevenue DESC;
```

Result Set

Product	Category	Revenue (KES)	Units Sold	Group
Trench Coat	Jackets	38,500	7	Top 10
Running Sneakers	Footwear	37,800	9	Top 10
Leather Handbag	Accessories	24,500	7	Top 10
Denim Jacket	Jackets	22,800	6	Top 10
Floral Maxi Dress	Dresses	22,500	9	Top 10
Block Heel Sandals	Footwear	19,200	6	Top 10
Bodycon Dress	Dresses	16,800	6	Top 10
Pleated Midi Skirt	Skirts	15,200	8	Top 10
Yoga Pants	Activewear	13,200	6	Top 10
Slim Fit Chinos	Trousers	12,600	7	Top 10
Linen Palazzo Pants	Trousers	2,100	1	Bottom 10
Beaded Bracelet	Accessories	3,600	8	Bottom 10
Graphic Tee	Tops	5,400	6	Bottom 10
Sports Bra	Activewear	5,600	4	Bottom 10
Printed Scarf	Accessories	5,850	9	Bottom 10
Cargo Shorts	Trousers	6,400	4	Bottom 10

Gold Hoop Earrings	Accessories	7,650	9	Bottom 10
Ankara Wrap Skirt	Skirts	10,500	7	Bottom 10
Canvas Backpack	Accessories	11,200	4	Bottom 10
Classic White Shirt	Tops	12,000	10	Bottom 10

Insight

The Trench Coat and Running Sneakers are the top revenue drivers at KES 38,500 and KES 37,800, driven by premium pricing rather than volume. The Classic White Shirt is the most sold product at 10 units but sits in the bottom 10 due to its low unit price of KES 1,200. This is a pricing problem, not a demand problem. Recommendation: review pricing on high-demand low-price items, protect the premium product range, and develop bundling strategies that pair entry-level accessories with premium products to increase average order value.

Query 4 Revenue by Product Category

Business Question

How is total revenue distributed across ShopZetu product categories? Which categories are driving the most value?

SQL Query

```
WITH ProductRevenue AS (
  SELECT
    P.category,
    SUM(OI.quantity * OI.unit_price) AS TotalRevenue
  FROM Order_Items OI
  INNER JOIN Products P ON OI.product_id = P.product_id
  GROUP BY P.category
)
SELECT
  category,
  TotalRevenue,
  ROUND(TotalRevenue * 100.0 / SUM(TotalRevenue) OVER(), 2) AS PercentageShare
FROM ProductRevenue
ORDER BY TotalRevenue DESC;
```

Result Set

Category	Total Revenue (KES)	Share %
Jackets	61,300	20.89%
Footwear	57,000	19.43%
Accessories	52,800	18.00%
Dresses	39,300	13.39%
Skirts	25,700	8.76%
Trousers	21,100	7.19%
Activewear	18,800	6.41%
Tops	17,400	5.93%

Insight

Jackets, Footwear, and Accessories account for 58.3% of total revenue, driven by premium pricing. Tops is the weakest category at 5.93% despite containing the most sold product in the dataset. This confirms a pricing problem in that category, not a demand problem. Activewear shows internal inconsistency where Yoga Pants ranks in the top 10 products while Sports Bra ranks in the bottom 10. Recommendation: review pricing on Tops, replace or reprice the Sports Bra, and monitor category concentration to avoid over-reliance on Jackets and Footwear.

Query 5 Repeat Purchase Rate

Business Question

What percentage of ShopZetu customers have made more than one purchase? How strong is customer loyalty?

SQL Query

```
WITH CustomerOrders AS (
  SELECT
    customer_id,
    COUNT(*) AS TotalOrders
  FROM Orders
  GROUP BY customer_id
)
SELECT
  COUNT(*) AS TotalCustomers,
  SUM(CASE WHEN TotalOrders > 1 THEN 1 ELSE 0 END) AS RepeatCustomers,
  ROUND(SUM(CASE WHEN TotalOrders > 1 THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS
RepeatRate
FROM CustomerOrders;
```

Result Set

Total Customers	Repeat Customers	Repeat Rate
20	14	**70.00%**

Insight

70% of ShopZetu customers are repeat buyers, nearly double the 40% benchmark set at the start of this analysis and a strong indicator of customer loyalty. However, 6 customers have purchased only once and represent an immediate re-engagement opportunity. Combined with zero new customer acquisition since October 2023, the business is entirely dependent on its existing 20-customer base. If even 3 to 4 of the 14 repeat buyers reduce purchasing frequency, revenue will decline with no new customers to absorb the loss. Recommendation: launch a win-back campaign targeting single-purchase customers and prioritise new customer acquisition as the most urgent growth lever.

Query 6 Average Days Between First and Second Purchase

Business Question

For customers who have made at least two purchases, how many days passed between their first and second order? What does this tell us about the re-engagement window?

SQL Query

```
WITH CustomerOrders AS (
  SELECT customer_id, order_date, COUNT(*) AS TotalOrders
  FROM Orders
  GROUP BY customer_id, order_date
),
OrderDates AS (
  SELECT
    C.full_name,
    CO.order_date,
    COUNT(CO.TotalOrders) OVER (PARTITION BY C.customer_id) AS TotalOrdersPlaced,
    LEAD(CO.order_date) OVER (PARTITION BY C.customer_id ORDER BY CO.order_date) AS
NextOrderDate,
    DENSE_RANK() OVER (PARTITION BY C.customer_id ORDER BY CO.order_date) AS OrderRank
  FROM Customers C
  INNER JOIN CustomerOrders CO ON C.customer_id = CO.customer_id
)
SELECT
  full_name,
  order_date AS FirstOrderDate,
  DATEDIFF(DAY, order_date, NextOrderDate) AS DaysBetweenOrders
FROM OrderDates
WHERE OrderRank = 1 AND TotalOrdersPlaced > 1;
```

Result Set

Customer	First Order Date	Days to Second Order
Amina Wanjiru	2023-01-15	66
Brian Otieno	2023-01-20	81
Carol Muthoni	2023-02-10	104
David Kiprop	2023-02-28	136
Esther Auma	2023-03-15	99
Felix Mutua	2023-04-05	136
Grace Chebet	2023-04-18	162
Irene Wambui	2023-05-20	117
Kevin Achieng	2023-06-15	129
Moses Njoroge	2023-07-10	138
Oscar Mwenda	2023-08-20	163
Purity Wanjiku	2023-09-10	86
Rose Kamau	2023-10-08	73

Tabitha Akinyi	2023-11-01	111

Insight

The average gap between first and second purchase is 114 days, just under 4 months. Customers vary from 66 days for the fastest return to 163 days for the slowest, suggesting a mix of frequent shoppers and more periodic buyers. Recommendation: trigger automated re-engagement campaigns at day 90 post-purchase for all customers, classify single-purchase customers as At Risk after 120 days of inactivity, and classify them as Lost after 180 days. Note: two data points per customer are not sufficient to confirm seasonal shopping patterns and more purchase history is needed before drawing those conclusions.

Query 7 RFM Customer Segmentation

Business Question

Based on Recency, Frequency, and Monetary value, how do ShopZetu customers segment? Who are the Champions, Loyal customers, At Risk, and Lost customers?

SQL Query

```
WITH CustomerMetrics AS (
  SELECT customer_id,
    COUNT(DISTINCT O.order_id) AS Frequency,
    SUM(OI.quantity * OI.unit_price) AS Monetary
  FROM Orders O
  INNER JOIN Order_Items OI ON O.order_id = OI.order_id
  GROUP BY customer_id
),
LastOrder AS (
  SELECT customer_id, MAX(order_date) AS LastOrderDate
  FROM Orders GROUP BY customer_id
),
RFM_Base AS (
  SELECT C.full_name,
    DATEDIFF(DAY, LO.LastOrderDate, '2024-08-01') AS Recency,
    CM.Frequency, CM.Monetary,
    NTILE(4) OVER (ORDER BY CM.Frequency ASC)                                  AS
NtileFrequency,
    NTILE(4) OVER (ORDER BY CM.Monetary ASC)                                   AS
NtileMonetary,
    NTILE(4) OVER (ORDER BY DATEDIFF(DAY,LO.LastOrderDate,'2024-08-01') DESC)   AS
NtileRecency
  FROM Customers C
  INNER JOIN CustomerMetrics CM ON C.customer_id = CM.customer_id
  INNER JOIN LastOrder LO ON C.customer_id = LO.customer_id
),
RFM_Scored AS (
  SELECT *, ROUND((NtileRecency+NtileFrequency+NtileMonetary)/3.0,2) AS AvgScore,
    CASE
      WHEN NtileRecency=4 AND NtileFrequency=4 AND NtileMonetary=4 THEN 'Champion'
      WHEN (NtileRecency+NtileFrequency+NtileMonetary)/3.0 >= 2.5  THEN 'Loyal'
      WHEN (NtileRecency+NtileFrequency+NtileMonetary)/3.0 >= 1.5  THEN 'At Risk'
      ELSE 'Lost'
    END AS Segment
  FROM RFM_Base
)
SELECT full_name, Recency, Frequency, Monetary,
  NtileRecency, NtileFrequency, NtileMonetary, AvgScore, Segment
FROM RFM_Scored
ORDER BY AvgScore DESC;
```

Result Set

Customer	Recency	Frequency	Monetary	R	F	M	Avg	Segment
Grace Chebet	73	4	26,300	4	4	4	4.00	**Champion**
Amina Wanjiru	18	7	43,650	4	4	4	4.00	**Champion**
Esther Auma	169	4	22,700	3	4	4	3.67	**Loyal**

Brian Otieno	199	4	25,900	2	4	4	3.33	**Loyal**
Tabitha Akinyi	54	3	16,900	4	3	3	3.33	**Loyal**
Moses Njoroge	108	3	21,500	3	3	4	3.33	**Loyal**
Purity Wanjiku	101	3	15,650	4	3	3	3.33	**Loyal**
Rose Kamau	83	3	15,800	4	3	3	3.33	**Loyal**
Kevin Achieng	155	3	15,600	3	3	3	3.00	**Loyal**
Irene Wambui	129	3	15,200	3	2	3	2.67	**Loyal**
Carol Muthoni	267	3	13,250	2	4	2	2.67	**Loyal**
Oscar Mwenda	184	2	12,600	3	2	2	2.33	**At Risk**
Felix Mutua	348	2	9,550	2	2	2	2.00	**At Risk**
Samuel Bett	291	1	5,000	2	2	1	1.67	**At Risk**
David Kiprop	384	2	8,600	1	2	2	1.67	**At Risk**
Henry Odhiambo	457	1	7,000	1	1	2	1.33	**Lost**
Quincy Omondi	311	1	5,200	2	1	1	1.33	**Lost**
James Kariuki	423	1	5,200	1	1	1	1.00	**Lost**
Lydia Rotich	397	1	4,500	1	1	1	1.00	**Lost**
Nancy Cherop	362	1	3,300	1	1	1	1.00	**Lost**

Segment Summary

Champion	Loyal	At Risk	Lost
2 customers	**9 customers**	**4 customers**	**5 customers**

Insight

RFM segmentation identified 2 Champions, 9 Loyal customers, 4 At Risk, and 5 Lost out of 20 total customers. Amina Wanjiru and Grace Chebet are the top performing customers. The 5 Lost customers have not purchased recently, buy infrequently, and spend the least. Note that RFM scores are relative, not absolute. In a base of 20 customers, the top 25% by frequency only requires 4 orders to qualify as a Champion. With zero new customer acquisition since October 2023, retaining the 9 Loyal customers is the single most important revenue protection priority. Recommendation: invest marketing budget in Champions and Loyal customers through loyalty rewards, target At Risk customers with win-back offers, and evaluate whether Lost customers are worth re-engagement cost given the absence of new acquisition to replace them.

Summary of Findings

#	Query	Key Finding	Recommendation
1	Monthly Revenue	Peak KES 24,700 in Jan 2024 then sharp decline	Verify data completeness for June and July 2024
2	Customer Acquisition	Zero new customers since October 2023	Target 5 new customers per month minimum
3	Product Performance	Classic White Shirt most sold but bottom 10 revenue	Review pricing on high-demand low-price products
4	Category Revenue	Top 3 categories account for 58.3% of revenue	Diversify category revenue and reprice Tops
5	Repeat Purchase Rate	70% repeat rate, nearly double the 40% benchmark	Win-back campaign for 6 single-purchase customers
6	Days Between Purchases	Average 114 days between first and second order	Trigger re-engagement at day 90 post-purchase
7	RFM Segmentation	2 Champions, 9 Loyal, 4 At Risk, 5 Lost	Protect Loyal base, evaluate Lost customer cost

End of Report | ShopZetu E-Commerce Analysis 2023 to 2024 | Benson Kioko